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1 Physical Proof of Existence

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- 2. *Yan Ju Yi Wang Gao* 偃矩以望高 and *Fu Ju Yi Ce Shen* 覆矩以测深 are the positive and negative*
- 3. *Wo Ju Yi Zhi Yuan* 卧矩以知远 is horizontal triangulation.*
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keywords:

- Zhoubi Suanjing
- Shang Gao
- Ju
- Groma
- Gui-Ju Dao
- Geometric Operator System
- 3D Orthogonal Frame
- Shen Zheng
- Wu Qi
- 4+1 Meta-Structure
- I Ching
- Cosmic Engineering

Groma--Ju Dao: The Six Modes of the *Zhoubi Suanjing* "Ju" as a Geometric Operator System

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Abstract

The opening dialogue of the *Zhoubi Suanjing* records Shang Gao's six oral instructions on the This paper presents a reconstructive thought experiment: **the six instructions are not independent**. Within this reconstruction we show that:

1. *Ping Ju Yi Zheng Sheng* 平矩以正绳 is the initialization of the 3D orthogonal frame $\{\mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z\}$
2. *Yan Ju Yi Wang Gao* 偃矩以望高 and *Fu Ju Yi Ce Shen* 覆矩以测深 are the positive and negative
3. *Wo Ju Yi Zhi Yuan* 卧矩以知远 is horizontal triangulation.
4. *Huan Ju Yi Wei Yuan* 环矩以为圆 is the rotation $R_Z(\phi)$ about the Z -axis, generating the
5. *He Ju Yi Wei Fang* 合矩以为方 is the translation combination (T_x, T_y) , generating orthogonal

The six operations unify under a single 3D orthogonal frame $\{\mathbf{F}\}_0$ as a three-level system.

Keywords: *Zhoubi Suanjing*; Shang Gao; Ju; Groma; Gui-Ju Dao; geometric operator system;

Notation

Symbol	Meaning	First Definition
$\{\mathbf{F}\}_0 = (0; \mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$	Observer-centered 3D orthogonal	
$\mathbf{O} = (0, 0, 0)^T$	Origin	Eq. (4)
Π_h	Horizontal reference plane	§1
$Z \parallel \mathbf{g}$	Vertical axis parallel to gravity	Eq. (3)
$R_Y(\theta)$	Rotation about Y -axis	Eq. (7)
$R_Z(\phi)$	Rotation about Z -axis	Eq. (15)
$T_x(b), T_y(b)$	Translation along X, Y -axes	Eqs. (11), (18)
$\{\mathbf{J}\}$	Six-mode operator system	Eq. (23)

1. Introduction: The Six Oral Instructions

The *Zhoubi Suanjing* opens with a dialogue:

> *The Duke of Zhou asked Shang Gao...* Shang Gao said: "Lay the Ju flat to align the plumb line

This paper does not address questions of textual dating, transmission, or cross-cultural influence.

We approach this as a **reconstructive thought experiment**. The experimental apparatus is defined by

\$\$

$$\boxed{\{\mathbf{S}\} = \{\text{Groma}, \text{Dioptra}, \text{Level}, \text{Plumb}, \text{Rota}, \text{Gui-Ju Dao}\}}$$

Component	Function	Geometric Role
Groma	Cross-frame with plumb lines, establishing straight lines and right angles	Orthogonal
Dioptra	Rotatable sighting tube, reading elevation/depression/horizontal angles	Angular

****Libra / Chorobates****	Leveling instrument	Horizontal 基准 (Π_h)
****Plumb****	Plumb line	Vertical 基准 $(Z \parallel \mathbf{g})$
****Rods / Cords****	Measuring rods and ropes	Length scale (L)

Crucially, no single instrument bears the full functional burden. The Ju, in this reconstruction

The present study confirms the traditional attribution of these six instructions to Shang Gao :

2. Mode 1: *Ping Ju Yi Zheng Sheng* 平矩以正绳 — Frame Initialization

****Translation:**** Lay the Ju flat to align the plumb line.

****Gromaticus Operation:****

1. Set the Groma stand. Align the central plumb line's metal point with the ground station P_0

\$\$
 $O \downarrow P_0. \quad (2)$
 \$\$

2. Level the cross-frame using the level/plumb, ensuring

\$\$
 $X, Y \in \Pi_h. \quad (3)$
 \$\$

3. The plumb line establishes

\$\$
 $Z \parallel \mathbf{g}. \quad (4)$
 \$\$

****Geometric Result:****

\$\$
 $\boxed{\mathcal{F}_0 = (0; \mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)} \quad (5)$
 \$\$

with

$$\begin{aligned} &\mathbf{e}_x \perp \mathbf{e}_y, \mathbf{e}_y \perp \mathbf{e}_z, \mathbf{e}_z \perp \mathbf{e}_x, \quad (6) \end{aligned}$$

and

$$\mathbf{O} = (0, 0, 0)^T. \quad (7)$$

A complete three-dimensional orthogonal reference frame is established. The observer is positioned

Visual: Groma stand with cross-frame, leveled horizontal plane Π_h , and vertical plumb line.

3. Mode 2: *Yan Ju Yi Wang Gao* 偃矩以望高 — Elevation

Translation: Incline the Ju to look at heights.

Target: Measure the height $H=TB$ of a tower, where the base B is accessible but the top T is not.

Gromaticus Operation:

1. Groma provides the orthogonal reference $\mathcal{F}_0$ at station O , with known horizontal plane Π_0 .
2. Dioptra is mounted and rotated to elevation angle θ , sighting the tower top T .
3. The sight line OT forms angle θ with the horizontal plane.

Geometric Result:

Under the idealized level-ground reconstruction:

$$\tan \theta = \frac{H}{D}, \quad (8)$$

$$\boxed{H = D \tan \theta}. \quad (9)$$

****Classical Similar-Triangle Version:****

\$\$

$$\frac{H}{D} = \frac{h}{d} \quad \Longleftrightarrow \quad \boxed{H = \frac{Dh}{d}}. \quad (10)$$

\$\$

****Geometric Essence:**** This is the frame transformation

\$\$

$$\boxed{R_Y(+\theta) \mathcal{F}_0}. \quad (11)$$

\$\$

The horizontal axis is tilted upward. This is the ****elevation projection**** of the 3D frame.

****Visual:**** Right triangle with horizontal base D , vertical height H , elevation angle θ .

4. Mode 3: *Fu Ju Yi Ce Shen* 覆矩以测深 — Depression

****Translation:**** Invert the Ju to measure depths.

****Target:**** Measure the depth H of a well or vertical shaft.

****Gromaticus Operation:****

1. Groma provides the orthogonal reference $\mathcal{F}_0$ at the well edge.
2. Dioptra is rotated to depression angle $-\theta$, sighting the well bottom B .
3. The sight line forms angle $|\theta|$ with the horizontal plane.

****Geometric Result:****

Under the idealized level-ground reconstruction:

\$\$

$$\boxed{H = D \tan |\theta|}. \quad (12)$$

\$\$

****Geometric Essence:**** This is the frame transformation

\$\$

$$\boxed{R_Y(-\theta) \mathcal{F}_0}. \quad (13)$$

\$\$

Yan and Fu are **not** two different mathematical methods; they are the positive and negative of each other.

\$\$

$$\boxed{\text{Yan} = R_Y(+\theta), \text{Fu} = R_Y(-\theta)}. \quad (14)$$

\$\$

They are geometric mirror images of each other.

Visual: Right triangle with horizontal offset HD , vertical depth HD , depression angle θ .

5. Mode 4: *Wo Ju Yi Zhi Yuan* 卧矩以知远 — Horizontal Triangulation

Translation: Lay the Ju horizontally to know distances.

Target: Measure an inaccessible horizontal distance AB .

Gromaticus Operation:

1. Groma provides the orthogonal reference $\mathcal{F}_0$ at station O .
2. Establish a baseline $OO' = b$ along one axis:

\$\$

$$T_x(b)\mathcal{F}_0. \quad (15)$$

\$\$

3. Dioptra measures the horizontal angle $\phi_1 = \angle AOB$ to target A .
4. A second observation from station O' provides $\phi_2 = \angle A'O'B$.
5. Triangulation yields the inaccessible distance AB .

Geometric Result:

Using the law of sines:

\$\$

$$\boxed{AB = \frac{b \cdot \sin \phi_1}{\sin(\phi_1 + \phi_2)}}. \quad (16)$$

\$\$

Or, in the classical similar-triangle form:

$$\boxed{AB = b \cdot \frac{d}{c}}. \quad (17)$$

Geometric Essence: This is the frame transformation

$$\boxed{T_x(b) \mathcal{F}_0}, \quad (18)$$

followed by horizontal angular projection. The orthogonal frame is extended through baseline m

Visual: Two observation stations O, O' with baseline b , target A , and horizontal angl

6. Mode 5: *Huan Ju Yi Wei Yuan* 环矩以为圆 — Circle Generation

Translation: Revolve the Ju to form a circle.

Target: Generate a precise circle in physical space.

Gromaticus Operation:

1. Groma fixes O and maintains $Z \parallel \mathbf{g}$: $\mathcal{F}_0$.
2. Fix one measuring arm at length r (constant).
3. Rotate the arm about the Z -axis:

$$0 \leq \phi < 2\pi. \quad (19)$$

4. The endpoint traces

$$\mathbf{r}(\phi) = r \begin{pmatrix} \cos \phi \\ \sin \phi \\ 0 \end{pmatrix}. \quad (20)$$

Geometric Result:

$$\boxed{x^2 + y^2 = r^2}. \quad (21)$$

\$\$

****Geometric Essence:**** This is the frame transformation

\$\$

$\boxed{R_Z(\phi)\mathcal{F}_0}.$ \quad (22)

\$\$

This is the first operation that ****requires**** the Z -axis established in Mode 1. The circle is

****The Gui (compass/circle) is geometrically the trajectory of the Ju (square/frame) rotated about**

This establishes a clear hierarchy:

\$\$

$\boxed{\text{Ju} \rightarrow \text{Orthogonality} \rightarrow \text{Rotation} \rightarrow \text{Translation}}$

\$\$

****Visual:**** Groma at center O , rotating arm of length r , circular trajectory $x^2+y^2=r^2$

7. Mode 6: *He Ju Yi Wei Fang* 合矩以为方 — Grid Composition

****Translation:**** Combine Ju to form a square.

****Target:**** Establish a precise orthogonal square grid.

****Gromaticus Operation:****

1. Use Groma to establish the first right angle $X \perp Y$ at origin O : $\mathcal{F}_0$.
2. Translate along X :

\$\$

$T_x(a)\mathcal{F}_0.$ \quad (24)

\$\$

3. Translate along Y :

\$\$

$T_y(b)\mathcal{F}_0.$ \quad (25)

\$\$

4. Repeat to generate four corners A, B, C, D .

Geometric Result:

AB $\parallel$ CD, $\square$
AD $\parallel$ BC, $\square$
AB $\perp$ AD, $\square$ (26)

$\square$
 $\boxed{ABCD = \text{square / rectangle}}$. $\square$ (27)
 $\square$

Geometric Essence: This is the frame composition:

$\square$
 $\boxed{T_x(a)T_y(b)\mathcal{F}_0}$. $\square$ (28)
 $\square$

The local frame is replicated across space to generate a global orthogonal grid. This is the op

Visual: Four corners A, B, C, D forming a square/rectangle, with orthogonal axes X, Y .

8. Synthesis: The Six Modes as a Three-Level Operator System

Figure 1. The Six Modes of the Ju as Operations on a 3D Orthogonal Frame.

[Figure 1. The Six Modes of the Ju as Operations on a 3D Orthogonal Frame.](images/DW58553910,

The six operations form a coherent three-level structure.

8.1 Level I – Frame Initialization

$\square$
 $\boxed{\text{Ping}}$
 $\square$

$$\mathcal{F}_0 = (0; \mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z). \quad (29)$$

8.2 Level II – Frame Transformation

$$\boxed{\text{Yan}, \text{Fu}, \text{Wo}, \text{Huan}}$$

$$\boxed{R_Y(+\theta), R_Y(-\theta), T_x(b), R_Z(\phi)} \quad (30)$$

8.3 Level III – Frame Composition

$$\boxed{\text{He}}$$

$$\boxed{T_x(a)T_y(b)\mathcal{F}_0} \quad (31)$$

This yields the unifying operator sequence:

$$\boxed{\mathcal{F}_0 \xrightarrow{R_Y(+\theta)} \text{Elevation} \xrightarrow{R_Y(-\theta)} \dots}$$

Mode	Operation	Geometric Operator	Level
---	---	---	---
Ping 平	Lay flat + plumb	Initialize $\mathcal{F}_0$	I: Frame Creation
Yan 偃	Incline upward	$R_Y(+\theta)$, $H=D\tan\theta$	II: Frame Transformation
Fu 覆	Invert downward	$R_Y(-\theta)$, $H=D\tan\theta$	II: Frame Transformation
Wo 卧	Lay horizontally	$T_x(b)$, triangulation	II: Frame Transformation

| **Huan** 环 | Revolve | $R_Z(\phi)$, $x^2+y^2=r^2$ | II: Frame Transformation |
 | **He** 合 | Combine | $T_x(a)T_y(b)$, orthogonal grid | III: Frame Composition |

The six instructions can thus be compressed into a single system:

\$\$
 \boxed{\mathcal{J} = \{\text{Initialize}, \text{Elevate}, \text{Depress}, \text{Project}\},
 \$\$

Each mode is a specific **projection, rotation, or composition** of the same $\mathcal{F}_0$ for

Figure 2. Schematic Reconstruction of the Roman Groma-Ju Discovered in Pompeii; not to scale

[Figure 2. Schematic Reconstruction of the Roman Groma-Ju Discovered in Pompeii; not to scale]

9. The *Gui-Ju Dao* as a Complete Cosmic Engineering System

9.0 From Operational Geometry to Ontological Symbolization

The six-mode Groma-Ju system is first an operational geometry. Yet any repeatedly executed oper

9.1 The Ultimate Goal: Measuring Heaven and Earth

The *Zhoubi Suanjing* does not present the six Ju operations as mere land-surveying tricks. Its

商高曰: "方属地, 圆属天, 天圆地方, 方数为典, 以方出圆。"

Shang Gao said: "The square pertains to Earth; the circle to Heaven. Heaven is round, Earth is

This is the foundational axiom of the entire *Gui-Ju Dao*: **the discrete, measurable square (I**

9.2 The *Zhoubi Suanjing* in Action: The *Kao Gong Ji* Record of State Surveying

The six-mode Groma-Ju reconstruction presented above finds a remarkable independent confirmati

The *Kao Gong Ji* records the complete surveying protocol for the foundation of a royal capita

> **匠人建国, 水地以县, 置槊以县, 视以景, 为规, 识日出之景与日入之景, 昼参诸日中之景, 夜考之极星, 以正朝夕。**

The Master Builder, when establishing a capital, levels the ground with a plumb line; sets up

This passage, when read through the Groma-Ju geometric operator system, reveals a complete sur

9.2.1 The Eight-Step Protocol as Groma-Ju Operations

Step	Text	Groma-Ju Operation	Geometric Function
------	------	--------------------	--------------------

	---		---		---		---	
	1		** 水地以县 **		(*level the ground with a plumb line*)		**Ping** 平	— frame initialization
	2		** 置槲以县 **		(*set up a gnomon with a plumb line*)		**Ping** 平	— vertical calibration
	3		** 视以景 **		(*observe its shadow*)		**Yan** 偃	— elevation observation Begin astronomi
	4		** 为规 **		(*make a circle*)		**Huan** 环	— circle generation $R_Z(\phi)$, $x^2+y^2=r^2$
	5		** 识日出之景 **		(*mark the sunrise shadow*)		**Wo** 卧	— horizontal projection Mark poi
	6		** 识日入之景 **		(*mark the sunset shadow*)		**Wo** 卧	— horizontal projection Mark poin
	7		** 昼参诸日中之景 **		(*consult the midday shadow*)		**He** 合	— grid composition Establis
	8		** 夜考之极星 **		(*examine the Pole Star at night*)		**Ping** 平	— frame verification Ca

9.2.2 The Central Operation: *Wei Gui* 为规 as *Huan Ju* 环矩

The fourth step, ** 为规 ** (*wei gui*, "make a circle"), is the geometric key to the entire pr

> ** 为规 is the rotation of the Groma cross-arm about the vertical Z -axis—the physical execu

The operation proceeds as follows:

1. The Groma is fixed at station O , with the plumb line ensuring $Z \parallel \mathbf{g}$.
2. One arm of the cross-frame is fixed at length r .
3. The arm is rotated about the Z -axis:

$$0 \leq \phi < 2\pi. \quad (34)$$

4. The endpoint traces the circle:

$$\boxed{x^2+y^2=r^2}. \quad (35)$$

**The circle is not drawn with a separate compass; it is generated by the rotation of the Ju-d

9.2.3 The Complete Geometric Protocol

When the full eight-step protocol is executed, the geometric results unfold as follows:

Steps 1–2 (Ping): The Groma establishes $\mathcal{F}_0 = (0; \mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z)$

Step 3 (Yan): The observer sights the sun, recording the shadow direction and length.

Step 4 (Huan): The Groma arm rotates, generating the circle

$$x^2+y^2=r^2 \quad (36)$$

on the ground.

Steps 5-6 (Wo): At sunrise, the shadow tip falls on the circle at $P_{\{\text{dawn}\}}$. At sun

$$\boxed{X_{\text{-axis}} = \overline{P_{\{\text{dawn}\}}P_{\{\text{dusk}\}}}} \quad (37)$$

Step 7 (He): At midday, the shortest shadow indicates the north-south direction, completing

$$\boxed{X \perp Y}. \quad (38)$$

Step 8 (Ping, verification): At night, the Pole Star confirms the north-south alignment, ca

9.2.4 The Cross-Civilizational Convergence

Operation	*Kao Gong Ji* (Zhou China)	Roman Gromaticus	Geometric Function
---	---	---	---
Level + plumb	水地以县 + 置槲以县	Level + Plumb	Establish Π_h , Z
Circle generation	为规	Groma rotation	$R_Z(\phi)$, $x^2+y^2=r^2$
East-west axis	识日出/日入之景	Decumanus (east-west)	X -axis from shadow points
North-south axis	昼参日中之景	Cardo (north-south)	Y -axis from midday shadow
Celestial calibration	夜考之极星	Pole Star sighting	Verify Y -axis

Two civilizations, separated by geography and language, confronting the same engineering problem.

9.2.5 The *Kao Gong Ji* as Independent Evidence

First, the *Kao Gong Ji* provides **independent documentary evidence** that the six-mode operation

Second, the *Kao Gong Ji* confirms the **central role of the circle** in the Ju-based surveying

9.2.6 A Note on Historical Interpretation

The *Kao Gong Ji* dates from the Warring States period (c. 5th-3rd century BCE), several centuries

The present study does not claim that the Warring States author of the **Kao Gong Ji** understood

9.2.7 Summary: The **Kao Gong Ji** as a Field Manual

The **Kao Gong Ji** "Master Builder" protocol can now be understood as a **field manual** for the

- The **level** and **plumb** establish the orthogonal frame $\mathcal{F}_0$.
- The **gnomon** and **shadow observations** project celestial information onto the ground.
- The **circle** generated by rotation about Z provides the geometric reference for determining
- The **midday shadow** and **Pole Star** confirm the north-south axis.
- The complete frame (X,Y,Z) provides the foundation for city planning—the "square" 方 of the

This protocol is, in effect, the **same geometric instruction set** that the Groma-Ju reconstructed

\$\$

\boxed{\text{The Kao Gong Ji records the Groma-Ju system in practice.}} \quad (39)

\$\$

\$\$

\boxed{\text{The six modes of Shang Gao are the instruction set; the Master Builder's protocol

\$\$

9.3 The Units of Measurement: Li, Bu, Du

The **Zhouli Suanjing** established a complete three-tier system of measurement units, enabling

- **Li 里**: the primary unit of terrestrial distance, used for mapping the extent of lands and
- **Bu 步**: the subunit of the Li, used for finer terrestrial measurements.
- **Du 度**: the unit of celestial arc, with a fixed conversion to linear distance—one degree

This system ensured that angular measurements of the sky could be converted into linear distance

9.4 The Professional Surveyors of the Zhou: Evidence from the **Zhou Li**

The **Zhouli Suanjing** was not an isolated text. The **Zhou Li** records a fully institutionalized

- **Liang Ren 量人**: in charge of establishing the methods for the capital cities and the nine
- **Tu Fang Shi 土方氏**: in charge of the gnomon (**Tu Gui**) method for establishing states and
- **Si Xian 司险**: in charge of roads and passes.
- **Si Kong 司空**: in charge of measuring land and distances.
- **Xing Fang Shi 形方氏**: in charge of the geographical forms of the states.

These offices confirm that the **Gui-Ju Dao** was not a philosophical abstraction but a **state-**

9.5 Conjecture: The Earliest Chinese Topographic Maps [L3]

The earliest known Chinese topographic map is the silk *Terrain Map* (*Dixing Tu*) unearthed from the Han Dynasty.

****Conjecture [L3]:**** This mapping capability must have relied on a mature geometric surveying system.

9.6 Conjecture: The Multiplication Table as a Mental Arithmetic Accelerator [L3]

The earliest known Chinese multiplication table is the *Jiujiu Biao* 九九表 inscribed on wooden slips from the Han Dynasty.

****Conjecture [L3]:**** The *Jiujiu Biao* served as a ****mental arithmetic accelerator**** for surveyors and administrators.

Linear Matrix Transformation of the *Jiujiu Biao* Arithmetic Accelerator in Mode 4

Under Mode 4: *Wo Ju Yi Zhi Yuan* 卧矩以知远, the observer establishes a terrestrial measurement framework.

Let the observer origin be $(0,0,0)^T$. Let a calibrated base distance along the X -axis denote L .

In the field manual protocol, a miniature scale-invariant physical alignment framework ("small scale") is used.

$$\boxed{x_A = \frac{b \cdot d}{c}}. \quad (41)$$

The *Jiujiu Biao* functions as a discrete look-up table:

$$\boxed{\text{Jiujiu Biao: } (b,d) \mapsto b \cdot d \quad \text{then divide by } c}. \quad (42)$$

By matching the integer values of b and d on the wooden slips, the division phase is bypassed.

9.7 The "4+1" Meta-Structure: A Structural Pattern Across Classical Chinese Cosmic Frameworks

If Sections 2-8 reconstruct the instruction set, Section 9.7 asks why this instruction set shows this pattern.

Alongside the Hetu 河图, the Luoshu 洛书, the *I Ching* 易经, the *Tai Xuan Jing* 太玄经, and the Wuxing 五行.

$$\boxed{\text{4 external pairs} + \text{1 central self-anchoring point}}. \quad (43)$$

Framework	The "4+1" Pattern
Hetu 河图	Four generative pairs around central 5: (1,6), (2,7), (3,8), (4,9), with (5,10)
Luoshu 洛书	Four symmetric pairs around central 5: (1,9), (2,8), (3,7), (4,6), with 5 in center
I Ching 易经	Four paired trigrams (乾坤, 兑艮, 离坎, 震巽) around the Taiji 太极
Tai Xuan Jing 太玄经	Four external duals (1 6, 2 7, 3 8, 4 9) with 5 5 self-pairing
Wu Xing 五行	Four phases (木火金水) circulating around the central Earth (土), with generation and control
Groma-Ju (Six Operations)	Four external dual domains: Vertical (偃 覆), Horizontal E-W (左 右)

9.7.1 The "1": *Shen Zheng* 身正 and the Observer-Gnomon Unity

The "1" is not a physical object but a state of being—the state of *shen zheng* 身正, "body upright."

Shen zheng operates on three levels:

- **Physical:** The observer's spine becomes the gnomon 梲. When the plumb-line verifies the gnomon's verticality, the body is upright.
- **Operational:** Every subsequent measurement depends on *shen zheng*. The causal chain is: *Shen zheng* → *Gnomon upright* → *Shadow truthful* → *Four directions true*.
- **Ontological:** The observer must first rectify himself before he can rectify the cosmos. The body is the root.

The "1" is thus defined as the indivisible unity of observer and gnomon:

\$\$
\boxed{\text{自本} \rightarrow \text{梲} \quad (\text{Self} \rightarrow \text{Gnomon})} . \quad \text{身正} \rightarrow \text{梲正} \rightarrow \text{景正} \rightarrow \text{四方正} \\
\$\$

* 身者，本也 *—the body is the root. This arrow expresses the dynamic of *Tian Ren He Yi* 天人合一.

The full causal chain can now be stated:

\$\$
\boxed{\text{身正} \rightarrow \text{梲正} \rightarrow \text{景正} \rightarrow \text{四方正}} . \quad \text{Body upright} \rightarrow \text{Gnomon upright} \rightarrow \text{Shadow truthful} \rightarrow \text{Four directions true} \\
\$\$

Body upright → Gnomon upright → Shadow truthful → Four directions true.

Shen zheng is not merely the first step within the "4+1" system—it is the precondition for the entire system.

A vernacular expression independently preserves this operational truth. The Chinese saying "身正影直" (Body upright, shadow straight) captures the essence of the process.

The shadow of the gnomon is necessarily slanted at all times except noon—this is not a defect but a confirmation of the process.

"Fear" and "not fearing" divide along a single criterion: when the body (and thus the gnomon) is upright, the shadow is truthful, and the directions are true.

This is not to deny the moral reading of the proverb; it is to recognize that the proverb's logic is operational and ontological.

\$\$

\boxed{\text{身正不怕影子斜——非谓影不斜，谓身正者能读影中之正也。}} \quad (46)

\$\$

*The upright body fears not the slanted shadow—not because the shadow is not slanted, but because

This saying thus preserves, in everyday language, the first principle of Cosmic Engineering: the

9.7.2 The "4": Four External Dual Domains

From the unified "1" (observer-gnomon), four external dual domains unfold.

1. Vertical Domain (偃 覆)

Yan Ju yi wang gao 偃矩以望高, "incline the gnomon to observe height," and *Fu Ju yi ce shen* 覆

2. Horizontal East-West Domain (卧东 卧西)

Wo Ju yi zhi yuan 卧矩以知远, "lay the gnomon to know distance," contains within it a dual meas

- *Kao Gong Ji* 考工记: "为规, 识日出之景与日入之景"—Draw a circle, mark the shadows of sunrise and sunset
- *Zhoubi Suanjing* 周髀算经: "以日始出, 立表而识其晷。日入, 复识其晷。晷之两端相直者, 正东西也"—At sunrise, erect a gnomon and mark its shadow. At sunset, mark its shadow. The two ends of the shadows that are straight are the true East and West
- *Huainanzi* 淮南子: Records the refined double-gnomon, three-point alignment method for eliminating error

Wo Dong 卧东, observing the sunrise shadow, and *Wo Xi* 卧西, observing the sunset shadow, constitute

3. Horizontal North-South Domain (卧南 卧北)

The north-south axis is established through a dual verification mechanism:

- **Solar method (*Wo Nan*):** "昼参诸日中之景"—By day, reference the shadow at midday; the shortest shadow indicates the true North
- **Stellar method (*Wo Bei*):** "夜考之极星"—By night, examine the Pole Star; the Pole Star's position relative to the gnomon indicates the true North

These two methods, one solar and one stellar, constitute independent evidence chains for the south-north axis

4. Rotational Domain (环 合)

Huan Ju yi wei yuan 环矩以为圆, "rotate the gnomon to make a circle," and *He Ju yi wei fang* 合

9.7.3 Arithmetical Verification

The structure can be expressed as

\$\$

$\boxed{1 \text{ (身正/平)} + 4 \text{ (dual domains)} \times 2 \text{ (directions)} = 9 \text{ (operational degrees of freedom)}}$

The six oral instructions (textual layer) logically expand into nine operational degrees of freedom.

" 夫道术，言约而用博者，智类之明。问一类而以万事达者，谓之知道。 "

*In the Dao of methods, concise words with broad application is the brightness of intellectual methods.

9.7.4 Cross-Textual Consistency

The east-west / north-south dual structure within *Wo Ju* is confirmed by five independent textual references.

1. ***Kao Gong Ji***: " 识日出之景与日入之景" (E-W) + " 昼参诸日中之景，夜考之极星" (N-S)
2. ***Zhoubi Suanjing***: " 晷之两端相直者，正东西也。中折之指表者，正南北也"
3. ***Huainanzi***: Double-gnomon three-point alignment for E-W + midpoint bisection for N-S
4. **Zheng Xuan's commentary on *Zhou Li***: " 将以正四方也"
5. ***Yingzao Fashi*** 营造法式: Preserves the complete E-W/N-S two-step structure

9.7.5 The Groma-Ju as Physical Instantiation

The Groma-Ju system does not exhaust the full "four external pairs" seen in the other frameworks.

The observer-gnomon unity finds a parallel in the Roman Groma: the operator stands beside the gnomon.

9.7.6 Linguistic Fossils: " 四平八稳" and " 身正不怕影子斜"

Two Chinese expressions independently preserve the "4+1" structural logic in everyday language.

**" 四平八稳" (*sì píng bā wěn*, "four-level, eight-stable")

Element	Meaning	Structural Role
---	---	---
**" —"	(implied center)	The stable core Central self-anchoring point (observer-gnomon unity)
**" 四平"	Four level directions	Four external dual domains
**" 八稳"	Eight stable directions	The four dual domains extended into full directional space

Together, the " —" (the implied center), the " 四平" (the four balanced directions), and the " 八稳" (the eight stable directions) form a complete structural logic.

**" 身正不怕影子斜" (*shēn zhèng bù pà yǐngzi xié*, "an upright body fears not the slanted shadow")

Element	Meaning	Structural Role
---	---	---
身正	(upright body)	Observer-gnomon unity The "1"—precondition for all measurement
影子斜	(slanted shadow)	Shadow at non-noon hours The mechanism of information encoding
不怕	(fears not)	Ability to decode The observer's capacity to extract truth from slant

The first idiom encodes the **spatial architecture** of the system $(4+1 \rightarrow 8)$; the s

9.7.7 The First Principle of Cosmic Engineering

The *Gui-Ju Dao*—as a physical system for orienting, measuring, and projecting space—operates

We may now state the first principle of Cosmic Engineering with precision:

\$\$

$\boxed{\text{身正，而后臬正；臬正，而后四方正；四方正，而后天下平。}} \quad (48)$

\$\$

Body upright, then the gnomon upright; gnomon upright, then the four directions true; four di

\$\$

$\boxed{\text{一平生四对偶，四对偶展八方向，八方向成天下秩序。}} \quad (49)$

\$\$

One Ping generates four dual domains; four dual domains unfold into eight directions; eight d

These two formulations are not separate principles but the same first principle stated at two l

\$\$

$\boxed{\text{The Gui-Ju Dao is a physical instantiation of the "4+1" meta-structure:}} \quad$

\$\$

\$\$

$\boxed{\text{one anchor, four projections, eight directions,}} \quad \text{and all under Heaven}$

\$\$

9.8 The Embodied Gnomon: Gregor Reisch's 1503 *Demonstratio* and the Cross-Cultural Alignm

9.8.1 The Mechanics of Flesh: The 1503 *Demonstratio* as Groma Embodiment

In Gregor Reisch's *Margarita philosophica* (1503), the woodcut illustration may be titled *Em*

Figure 3. Embodiment of the 3D Orthogonal Coordinate System. Gregor Reisch, *Margarita philos*

!Figure 3. Embodiment of the 3D Orthogonal Coordinate System. Gregor Reisch, *Margarita philos*

The Groma, the foundational hardware interface of Roman imperial engineering, relies on a hori

- **The Longitudinal Axis (Z-axis):** The Groma's vertical staff and plumb-line are absorbed by
- **The Transverse Axis (X-axis):** The left-to-right cross-arm of the *stella* is mapped onto
- **The Sagittal Axis (Y-axis):** The forward-to-backward sighting plane of the Groma is inter

Through this technological transubstantiation, the operator does not merely use an instrument;

\$\$

$\mathcal{F}_0 = (0; \mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z). \quad (52)$

\$\$

The intersection of these fleshy axes defines the geometric origin $O = (0, 0, 0)^T$, positioning t

The accompanying Latin text states: "*tribus se feri orthogonalis, ter in uno puncto secantibus*

9.8.2 Structural Invariance: The Geometry of Shang Gao's *Shen Zheng* 身正

This Western internalization of the Groma converges perfectly on the ontological and geometric

When Shang Gao demands that the observer stand upright to rectify the gnomon (表 or 臬), he is c

\$\$

$\boxed{\text{身正 (Body Upright)}} \rightarrow \text{臬正 (Gnomon Upright)}$

\$\$

In this configuration, the observer's body is not an external entity looking at a coordinate sy

9.8.3 The Convergence of the "4+1" Meta-Structure

The cross-cultural alignment between the 1503 Groma embodiment and Shang Gao's *Shen Zheng* can

Dimension / Operator	Reisch's 1503 <i>Demonstratio</i> (Western)	Shang Gao's Groma-Ju System
---	---	---
The "1" (Central Anchor)	Spinal alignment with gravity vector	<i>Shen Zheng</i> 身正 / Gnomon
Vertical Domain	Superior-Inferior Transfixion	<i>Yan Ju</i> 偃矩 <i>Fu Ju</i> 覆矩 Vertical A
Horizontal Planes	Left-Right / Front-Back Piercing	<i>Wo Ju</i> 卧矩 → Dawn-Dusk / Midday-S
Rotational/Grid Composites	Circumscribed Spatial Volumetrics	<i>Huan Ju</i> 环矩 <i>He Ju</i> 合矩

While the mathematical output is identical—the creation of a 3D orthogonal coordinate framewor

- **The Western Mode (Mechanical Penetration):** The 1503 woodcut utilizes a violent visual me
- **The Eastern Mode (Organic Resonance):** Shang Gao's *Shen Zheng* utilizes an ethical-cosmo

Ultimately, Reisch's *Homo Gromaticus* and Shang Gao's *Homo Ju-Dao* are the same mathematical

This cross-cultural alignment—between Reisch's pierced homunculus and Shang Gao's upright gnomon—

9.9 The *Wu Qi* 無器: The Origin as Vessel in the *Xuan Yin Yi Mi*

9.9.1 The *Wu* Segment: Origin as the Beginning and End of All Measure

The *Xuan Yin Yi Mi* 玄隱遺密, a Shang-era text orally transmitted for over three millennia and

```
> ** 大要以 Wu 為有之始終.**  
>  
> **Wu 器無間之容也.**  
>  
> ** 妙 Wu 有之玄。鼓吹萬物之榮。是為之紀.**
```

The character *Wu* (無本字) in its archaic form resembles an open vessel—suggesting not emptiness

****Translation and decoding:****

Phrase	Translation	Cosmic Engineering Decoding
大要以 Wu 為	** 有之始終 **	In sum, Wu is the beginning and end of all "having" (measure, manipulation)
**Wu 器	無間之容也	Wu is a vessel without gaps, that which contains without interruption.
妙 Wu 有之	** 玄 **。	鼓吹萬物之
	** 榮 **。	是為之
	** 紀 **。	The mystery of Wu is the mystery of "having"

9.9.2 The *Wu Qi* as the Conceptual Origin of the Jade Cong

The term **Wu Qi** 無器, "Wu-vessel," is the conceptual antecedent to the physical Jade Cong 玉琮

Wu Qi Attribute	Jade Cong Correlate	Cosmic Engineering Function
" 無間之容" (without gap, containing)	The hollow bore 中孔 of the Jade Cong	The \$Z\$-axis channel
" 有之始終" (beginning and end of all having)	The Jade Cong as the central hardware element	The origin \$O\$ as beginning and end
" 鼓吹萬物之榮" (breathes forth the flourishing of all things)	The Jade Cong's inner circle groove	The six directions from the origin, unifying
" 是為之紀" (this is its canon/rule)	The Jade Cong as the standard unit of measure	The *Ji* (measure)

9.9.3 The *Wu Qi* and the 1503 Woodcut: A Cross-Cultural Convergence

The 1503 Reisch woodcut—the human figure transfixed by three orthogonal axes—is a visual realization

Reisch Woodcut Element	*Wu Qi* Correlate	Cosmic Engineering Function
Human body as coordinate frame	"Wu 器無間之容"	The body as the continuous vessel containing
Intersection of three axes at the body's center	" 有之始終"	The origin \$O\$ as beginning and end
Six directions labeled on the figure	" 鼓吹萬物之榮"	The six directions from the origin, unifying
The figure as the *Homo Gromaticus*	" 是為之紀"	The observer as the living gauge (*Ji*) of

9.9.4 *Gui Yuan* 歸元: Return to the Origin as the Completion of Every Cycle

The *Xuan Yin Yi Mi* states:

> **" 物之安致，曰歸；用之通神，曰元。 "**

*Making things settle into their proper place is called Gui (return); using a tool to the point

In the Groma-Ju framework, this is the exact definition of the measurement cycle:

- **歸 (Gui):** The act of returning to the origin 0 after each measurement—what a modern engineer might call a "reset".
- **元 (Yuan):** The state of being aligned with the cosmic origin—what a modern engineer might call a "calibration".

Therefore:

$$\boxed{\text{归零} = \text{歸元}}. \quad (54)$$

The Gromaticus does not merely "zero" his instrument; he **returns to the origin** (*gui yuan*).

9.9.5 The *Xuan Yin Yi Mi* as the Missing Link

The *Xuan Yin Yi Mi* serves as the missing philosophical link between the practical operations

1. **The origin 0 is not a mathematical abstraction**—it is a *Wu Qi*, a continuous vessel that contains the three origins.
2. **The act of measurement is the act of manifesting the origin**—making the *Wu Qi* visible through the instrument.
3. **The observer's body is the ultimate *Wu Qi***—the living vessel that contains the three origins.
4. **Returning to the origin 归零/歸元 is not merely a technical reset**—it is the completion of the measurement cycle.

The *Wu Qi* is the origin as vessel; the Jade Cong is the *Wu Qi* in jade; the Groma is the *Wu Qi* in metal.

9.9.6 The Origin of Writing: The Character *Zhi* 知 as the Embodied Gnomon in Script

The *Xuan Yin Yi Mi* states:

" 纹之记曰文。无之记曰爻。形之附其图。象之会曰字。文始者 Zhi ('知'的象形字) 也。 "

Figure 4. The Pictograph *Zhi* 知 as an Embodied Gnomon and Projective Coordinate Frame.

[Figure 4. The Pictograph *Zhi* 知 as an Embodied Gnomon and Projective Coordinate Frame.] (image)

This passage reveals that the pictograph *Zhi* 知—meaning "to know"—originally depicted an observer using a gnomon to measure the sun's position.

This predates the 1503 Reisch woodcut by over two millennia, yet it encodes the same cognitive process of projective geometric mapping.

Projective Geometric Perspective Mapping of the Ideographic Character *Zhi* 知

The text of the **Xuan Yin Yi Mi** establishes that the archaic character 知 (**Zhi**—"to know") v

\$\$
\text{Observing Eye / Focal Node } E(0,0,h)
\$\$

\$\$
\downarrow \quad \text{(Projective Ray)}
\$\$

\$\$
\text{Plane } \Pi_c : \text{Circular Spatial Boundary}
\$\$

\$\$
\downarrow
\$\$

\$\$
\text{Ground Plane } \Pi_0 : \text{3D Orthogonal Grid}
\$\$

Let the continuous coordinate frame of Heaven and Earth be anchored at origin $0(0,0,0)^T$. The

The projection of any arbitrary spatial coordinate point $\mathbf{P}=(X,Y,Z)^T$ within the 3D c

\$\$
\boxed{ \mathbf{p} \sim \mathbf{P}_{\text{Zhi}} \mathbf{P} = \begin{pmatrix} f & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix} }
\$\$

Where $\mathbf{R}$ represents the $SO(3)$ rotation alignment matrix governed by the state of **Zhi**

****The Projective Invariants of the Character Geometry:****

1. The Upper Component (The Eye): Represents the focal origin $E(0,0,h)$ looking down along the

2. The Lower Component (The Grid): Represents the vanishing points of the parallel terrestrial
To "know" (*Zhi*) means, in the most foundational mathematical sense, to apply a non-degenerate

Mathematical Deconstruction of the Proto-Ideogram *Zhi* 知

The physical morphology preserved via the oral lineage of the *Xuan Yin Yi Mi* provides empirical

1. **The Superior Focal Domain $\mathcal{V}_{\text{eye}}$:**

The uppermost bounded elliptic curvature represents the eye of the observer, establishing the

2. **The Dynamic Transition Sector $\mathcal{V}_{\text{flow}}$:**

The intermediate divergent strands represent the non-linear transformation trajectories $\mathcal{M}$

3. **The Confined Orthogonal Subvariety $\mathcal{V}_{\text{base}}$:**

The base of the glyph displays a structurally pristine asymmetric round continuum boundary, sa

Thus, the ideographic construct of knowing (*Zhi*) is rigorously verified as a structural invar

\$\$

$\text{Zhi} \equiv \mathcal{V}_{\text{eye}} \xrightarrow{\hat{\mathbf{P}}_{\text{Zhi}}} \mathcal{M}$

\$\$

The glyph mathematically locks the observer, the information flow, and the cosmic coordinate g

9.9.7 The Sixty-Four Hexagrams: The Complete Symbolic System of Cosmic Engineering

The *Xuan Yin Yi Mi* states:

> **義王圖作，六十有四文。文天地，字人物。**

*King Xi (Fu Xi) composed the diagrams—sixty-four *wen* (patterns). The *wen* encode Heaven and

This passage reveals that the sixty-four hexagrams of the *I Ching* are not divinatory symbols

\$\$

$64=2^6$. \quad (57)

\$\$

| Factor | Cosmic Engineering Interpretation |

| --- | --- |

| ****6**** | The six modes of the Ju: Ping, Yan, Fu, Wo, Huan, He |

| ****2**** | The binary opposition inherent in each mode: positive/negative, up/down, front/back, |

| 2^6 | All possible combinations of the six operations and their opposites |

The six hexagram lines 六爻 correspond directly to the six Ju operations:

Hexagram Line	Ju Operation	Geometric Function
---	---	---
**Line 1 初爻 **	Ping 平 — Frame Initialization	Establish $\mathcal{F}_0=(0;X,Y,Z)$
**Line 2 二爻 **	Yan 偃 — Elevation	$R_Y(+\theta)$, upward projection
**Line 3 三爻 **	Fu 覆 — Depression	$R_Y(-\theta)$, downward projection
**Line 4 四爻 **	Wo 卧 — Horizontal Triangulation	$T_x(b)$, horizontal extension
**Line 5 五爻 **	Huan 环 — Circle Generation	$R_Z(\phi)$, rotational closure
**Line 6 上爻 **	He 合 — Grid Composition	T_xT_y , orthogonal grid generation

The binary opposition in each line (Yin/Yang, broken/unbroken) corresponds to the positive/negative

Line	Yang (—)	Yin (--)
---	---	---
Line 1	Origin established	Origin unestablished
Line 2	Elevation upward $(+\theta)$	Depression downward $(-\theta)$
Line 3	Forward projection	Backward projection
Line 4	Rightward extension	Leftward extension
Line 5	Clockwise rotation	Counterclockwise rotation
Line 6	Grid completed	Grid open

**The sixty-four hexagrams are thus the complete operational state space of the Groma-Ju system

The passage further distinguishes two functions of this symbolic system:

- ** 文天地 ** (*wen Heaven and Earth*): The *wen* (patterns/lines) encode the order of Heaven
- ** 字人物 ** (*zi persons and things*): The *zi* (characters/naming) name persons (the observed)

The *Xuan Yin Yi Mi* thus provides the ontological grounding for the *I Ching*: the sixty-four

\$\$

\boxed{ \text{The sixty-four hexagrams = the complete operational state space of the Groma-Ju system} }

\$\$

**The I Ching is not a divination manual; it is the cosmic engineering algorithm encoded in binary

Lie Algebra Register for the $2^6=64$ Hexagram State Space: A Conjectural Operator Mapping

The sixty-four hexagrams of the *I Ching* represent the discrete state space of the Groma-Ju operations

\$\$

$\mathcal{H} \in \{0,1\}^6 \cong \mathbb{Z}_2^6. \quad (59)$

\$\$

We propose a formal registration of the $2^6=64$ hexagram states into the discrete orbit genera

Let a general continuous state in the physical manifold be mapped to the Lie algebra $\mathfrak{so}(6)$

\$\$

$J_x = \begin{pmatrix} 0 & 0 & 0 \\ 0 & 0 & -1 \\ 0 & 1 & 0 \end{pmatrix}, \quad J_y = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ -1 & 0 & 0 \end{pmatrix}, \quad J_z = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & -1 \end{pmatrix}$

\$\$

The 6 modes of Shang Gao are mapped to a continuous projection operator $\hat{\Pi}_{\text{Ju}}$

To formalize the registration of the $2^6=64$ combinatorial states, we construct the Geometric

\$\$

$$\mathbf{\Omega}_k = \bigotimes_{\ell=1}^6 \left[y_{\ell} \mathbb{I}_3 + (1-y_{\ell}) \mathbb{I}_3 \right]$$

\$\$

Where $y_{\ell} \in \{0,1\}$ dictates the binary status of the line (0 for Yin, 1 for Yang), and $\mathbf{\Omega}_k$

\$\$

$\alpha(1)=z \ (\text{Ping}), \quad \alpha(2)=y \ (\text{Yan}), \quad \alpha(3)=y \ (\text{Fu}), \quad \alpha(4)=x \ (\text{Wo}), \quad \alpha(5)=z \ (\text{Huan}), \quad \alpha(6)=x,y \ (\text{He}).$

quad (62)

\$\$

When a hexagram state transitions (e.g., from Qian to Kun), the operator $\mathbf{\Omega}_k$ e

9.9.8 The Geometric Axiom: " 周回一也，方道四而八"

The *Xuan Yin Yi Mi* states:

" 天象與圓，地道行方。圓曰周回，方曰方道。周回一也，方道四而八。……周而复，无已故。"

This passage constitutes the most concise geometric axiom of the entire *Gui-Ju Dao*:

- **Heaven is round 圓:** circular, continuous, returning to itself—corresponding to *Huan Ju*
- **Earth is square 方:** discrete, gridded, extendable in four directions—corresponding to *He Ju*
- **" 周回一也":** the circle returns to one—the central anchor, the origin 0 .

- **"方道四而八":** the square extends from four to eight—the four cardinal directions unfolding
- **"周而复，无已故":** the cycle returns and repeats without end—the measurement cycle of depart

This passage provides the textual source for the "1+4" meta-structure formalized in Section 9.

10. Discussion: The Ju as Operational Geometry and Its Cognitive Implications

The reconstructive thought experiment and the geometric deconstruction of the proto-ideogram *

10.1 The Universality of the "4+1" Meta-Structure

The convergence between Shang Gao's six oral instructions, the Western Roman Gromaticus protoc

10.2 The Ontological Grounding of the *Wu Qi* 無器

Under this framework, the mathematical origin $(0,0,0)^T$ ceases to be an inert, abstract not

- **In Material Culture:** The central hollow bore of the Neolithic Jade Cong materializes this
- **In Bodily Rectitude:** The principle of *Shen Zheng* 身正 transforms the observer's spine in
- **In Algorithmic Memory:** The sixty-four hexagrams of the *I Ching* represent the exhaustiv

10.3 The Metrology Cycle as Cosmic Return

Finally, the engineering action of "zeroing" an instrument is elevated to its ontological root

11. Conclusion

This paper has presented a reconstructive thought experiment: treating the six oral instruction

We have demonstrated within this reconstruction that:

1. *Ping Ju Yi Zheng Sheng* 平矩以正绳 initializes the 3D orthogonal frame

\$\$

$\mathcal{F}_0 = (0; \mathbf{e}_x, \mathbf{e}_y, \mathbf{e}_z). \quad (63)$

\$\$

2. *Yan Ju Yi Wang Gao* 偃矩以望高 and *Fu Ju Yi Ce Shen* 覆矩以测深 are the positive and negative
3. *Wo Ju Yi Zhi Yuan* 卧矩以知远 is horizontal triangulation.
4. *Huan Ju Yi Wei Yuan* 环矩以为圆 is the rotation $R_Z(\phi)$ generating the circle $x^2+y^2=r^2$
5. *He Ju Yi Wei Fang* 合矩以为方 is the translation combination (T_x, T_y) generating orthogona

The six operations unify under a single 3D orthogonal frame $\mathcal{F}_0$ as a three-level sy

Methodological Output

This paper is not merely an exercise in using modern mathematics to interpret an ancient text. The six oral instructions of Shang Gao are not best understood as isolated measurement tricks, The broader lesson is this: ancient civilizations did not lack abstraction. They encoded abstraction. The value of this reconstruction lies not in claiming cross-cultural transmission, but in revealing it. The *Xuan Yin Yi Mi* supplies the ontological ground: the origin is the *Wu Qi*, the vessel that

$$\boxed{\text{玉琮载道, Groma 行之, 周髀算之, 身正歸元。}} \quad (64)$$

$$\boxed{\text{The Jade Cong bears the Dao; the Groma performs it; the Zhoubi computes it; the vessel}}$$

One is the cosmos rendered as form; the second, the cosmos rendered as action; the third, the cosmos rendered as measurement. The *Zhoubi Suanjing* gave us its language; the Groma gave us its hands; the *Xuan Yin Yi Mi* gave us its

Acknowledgments

We are grateful to all the individual scientists and teams of scientists who have contributed to this work.

Data Availability Statement

Code, mathematical scripts, and preprints are available at GitHub: <https://github.com/Zaitian001>

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1.1 Physical Proof of Existence (Appendix)

- **Banknote Serial:** DW58553910
- **SHA-256 Hash:** c8d1da1032e277e8fc065b2b73211a8bdbaa02225960b12fba3b82f52b71b232
- **Verification URL:** <https://Zaitian001.github.io/-Self-Preprint-V1.1-/preprints/DW58553910.html>

This document was timestamped via the Self-Preprint system.